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Ans(01)-

Redux is a state management library used in React to manage global application state in predictable way.

- Store → holds the state.
- action → object that describes what happened.
- Reducer → function that updates state based on action.
- Dispatch → sends action to reducer.

Benefits -

- Centralized State.
- Easy debugging.
- Predictable data flow.

Ans(02)-

Install Redux and React-Redux;
npm install redux react-redux

ex.

```
import { createStore } from "redux";  
const initialState = {  
  count: 0  
};  
function counterReducer(state = initialState, action) {  
  switch (action.type) {  
    case "INCREMENT":  
      return { count: state.count + 1 };  
    case "DECREMENT":  
      return { count: state.count - 1 };  
    default:  
      return state;  
  }  
}  
const store = createStore(counterReducer);  
export default store;
```

```

ex.
import React from "react";
import ReactDOM from "react-dom/client";
import { Provider } from "react-redux";
import App from "../app";
import store from "../store";

const root = ReactDOM.createRoot(
  document.getElementById("root")
);

root.render(
  <Provider store={store}>
    <App />
  </Provider>
);

```

Ans (03) -

```

ex.
function App() {
  const count = useSelector((state) => state.count);
  const dispatch = useDispatch();
  return (
    <div>
      <h2> Count: {count} </h2>
      <button onClick={() => dispatch(
        { type: "INCREMENT" }
      )}>
        Increment
      </button>
      <button onClick={() => dispatch(
        { type: "DECREMENT" }
      )}>
        Decrement
      </button>
    </div>
  );
}

export default App;

```